1. Professional Headline

PhD in Food Science | Non-Destructive Quality Assessment | NIR |Machine Vision| Laser light backscattering imaging| post-harvest | Food quality control| Food Processing

2. Summary

PhD in Food Sciences, specializing in non-destructive techniques for quality assessment of fruits and vegetables. My research focuses on NIR spectroscopy, machine vision, and postharvest quality evaluation. I am interested in research, innovation, and industry applications in food quality and postharvest technology.

3. Education

- PhD in Food Science, Hungarian University of Agriculture and Life Science, Budapest, Hungary, 2026
- MSc in Food Engineering, Addis Ababa Institute of Technology, Addis Ababa, Ethiopia, 2016
- BSc Chemical Engineering, Adama Science and Technology University, Adama, Ethiopia, 2013

4. Position Held

- Director: Beles Institute, Adigrat University, July 2019- 2021
- Research and Community Service Head: Beles Institute, Adigrat University, October 2017 to June 2019
- Food Engineering Team Coordinator, Department of Chemical Engineering, Adigrat University, October 2016 to September 2017
- Lecturer: Department of Chemical Engineering, Adigrat University, July 2016 – June 2021

5. Research interest/ Skill

- Non-destructive testing (Machine vision, NIR, Laser light backscattering imaging)
- Postharvest storage and quality assessment
- Food product development and characterization
- Data analysis (chemometrics, Octave, Minitab, R)

6. Publication(s)

Siyum, Z.H., Pham, T.T., Sao Dam, M., Friedrich, L.F., Hitka, G., Nguyen, L.L.P. and Baranyai, L., 2024. Assessment of plum quality changes during postharvest storage using multispectral NIR technique. *Journal of Agriculture and Food Research*, p.101476. https://doi.org/10.1016/j.jafr.2024.101476

Siyum, Z.H., Pham, T.T., Vozáry, E. et al. Monitoring of banana's optical properties by laser light backscattering imaging technique during drying. Food Measure (2023). https://doi.org/10.1007/s11694-023-02019-y

Siyum, Z.H., Pham, T.T., Nguyen, L.L.P., Baranyai, L., 2023a. Non-destructive monitoring of asparagus (Asparagus officinalis, L) quality changes during storage using NIR spectroscopy. Int J of Food Sci Tech ijfs.16704. https://doi.org/10.1111/ijfs.16704

Zinabu Hailu Siyum and Tassew Alemayehu Meresa (2021). Physicochemical Properties And Nutritional Values Of Carissa Spinarum L . /"Agam" Fruit International Journal of Fruit Science 21(1):826-834

Lien Le Phuong Nguyen ,Tung Thanh Pham , **Zinabu Hailu Syium** Viktória Zsom-Muha ,László Baranyai , Tamás Zsom and Géza Hitka (2022) Delay of 1-MCP Treatment on Post-Harvest Quality of 'Bosc Kobak' Pear Horticulturae 8(2):89 DOI: 10.3390/horticulturae802008

Pham, Thanh Tung, **Zinabu Hailu Siyum**, Thi Thanh Nga Ha, Hoa Xuan Mac, Sao Mai Dam, Trang Ha Dieu Nguyen, Lien Le Phuong Nguyen, and László Baranyai. Evaluating the quality of green asparagus treated with cassava starch-based coating using laser scattering. P*rogress in Agricultural Engineering Sciences* 19, no. S1 (2023): 59-68**.** **https://doi.org/10.1556/446.2023.00083**

Siyum, Z.H., Thanh, T.P., Höhn, M., Baranyai, L. and Nguyen, L.P.L., 2023, June. Monitoring of surface changes of fresh green asparagus during storage using laser light backscattering imaging. In VII International Symposium on Applications of Modelling as an Innovative

Technology in the Horticultural Supply Chain- 1382 (pp. 179-186).10.17660/ActaHortic.2023.1382.23

Gebreziher, Haftay Gebreyesus, Simon Zebelo, Yohannes Gerezihier Gebremedhin, Gebremedhin Welu Teklu, Yemane Kahsay Berhe, Daniel Hagos Berhe, Araya Kahsay Gerezgiher, Araya Kiros Weldetnsae, **Zinabu Hailu**, Gebrekidan Tesfay Weldeslasse, et al. 2025. Trends of Cochineal (Dactylopius coccus) Infestation as Affected by Armed Conflict, and Intervention Mechanisms for Sustainable Management in Tigray, Northern Ethiopia. Plants 14, no. 8: 1228. https://doi.org/10.3390/plants14081228

Shukla SP, **Hailu Z,** Kehsay Y. 2019. In vivo Rooting, Acclimatization and Ex situ Conservation of Opuntia ficus-indica (L.) Mill. (Beles) in Tigray Region of Ethiopia, Africa. International Journal of Plant and Environment . 5(04):265-9. : https://ijplantenviro.com/index.php/IJPE/article/view/1111

7. Award

Zinabu Hailu Siyum won the ISHS Prof. Jens Wünsche Young Minds Award for the best poster presentation at the International Symposium on Postharvest and Horticultural Products Quality at EHC2024 in Romania in May 2024 (**Monitoring changes in plum fruit quality during postharvest storage | International Society for Horticultural Science**)

Zinabu Hailu Siyum won the Best research paper for PhD student Award at WARESA Europe Award held on June 27, 2025 in Berlin, Germany

8. Professional Memberships

– International Society for Horticultural Science (ISHS)
– Association of Ethiopian Chemical Engineers (AEChE)

9. Contact and Links

ORCID ID: https://orcid.org/0000-0001-8601-0477

ResearchGate: https://www.researchgate.net/profile/Zinabu-Siyum

Google Scholar: Zinabu Hailu Siyum - Google Scholar

LinkedIn: https://hu.linkedin.com/in/zinabu-hailu-siyum-868369267

Email: zinabuhailiu84@gmail.com

10. References

Prof. József Felföldi
Hungarian University of Agriculture and Life Sciences, *Institute of Food Science and Technology*
Department of Measurements and Process Control, H-1118 Budapest, Somlói út 14-16.
Tel:+36/1/305-7620; E-mail: felfoldi.jozsef@uni-mate.hu

Dr. Salamon Bertold
MFP Vezető Technológus / MFP Lead Technical Manager, Tesco-Global Zrt.
H-2040 Budaörs, Kinizsi u. 1-3, Telefon: +36208273843
E-mail: Bertold.Salamon@tesco.com

Dr. Haftay Gebreyesus Gebreziher
Research, Publication and Ethics Director, Adigrat University
Tel: +251915707752, P.O.Box 50, Adigrat, Tigray, Ethiopia
Website: http://www.adu.edu.et Email; haftay.gebreyesus@gmail.com
ORCID: https://orcid.org/0000-0002-1070-7507
Researchgate: https://www.researchgate.net/profile/Haftay-Gebreziher